

Edexcel International A Level

Mathematics & Further Mathematics — Detailed Content Index (Y12 / Y13)

A chapter- and sub-section–level index of the Edexcel (Pearson) **International Advanced Level (IAL)** in Mathematics and Further Mathematics (2018 specification), aligned to the Pearson IAL student-book series used at international schools such as Cranleigh Abu Dhabi.

A Level Mathematics = P1, P2, P3, P4 + two applied units (here M1, S1). • **A Level Further Mathematics** = FP1, FP2, FP3 + applied units (here D1, M2, S2).

Year split (typical). Y12 (International AS) usually covers P1, P2 and one applied unit; Y13 (International A2) covers P3, P4 and a second applied unit. For Further Maths, FP1 and lower-numbered applied units sit in Y12, with FP2/FP3 and higher applied units in Y13. Exact sequencing varies by centre.

Accuracy. **VERIFIED** = confirmed against the Pearson IAL book contents page. **RESEARCHED** = chapter structure confirmed via research of the Pearson IAL books; sub-section titles within are best-effort and worth a final spot-check. The source PDFs cannot be opened directly from this environment, so verification was done through search.

Part 1 — Mathematics ("normal maths")

P1 — Pure Mathematics 1

VERIFIED

1 Algebraic expressions

- 1.1 Index laws
- 1.2 Expanding brackets
- 1.3 Factorising
- 1.4 Negative and fractional indices
- 1.5 Surds
- 1.6 Rationalising denominators

2 Quadratics

- 2.1 Solving quadratic equations
- 2.2 Completing the square
- 2.3 Functions
- 2.4 Quadratic graphs
- 2.5 The discriminant

3 Equations and inequalities

- 3.1 Linear simultaneous equations
- 3.2 Quadratic simultaneous equations
- 3.3 Simultaneous equations on graphs
- 3.4 Linear inequalities
- 3.5 Quadratic inequalities
- 3.6 Inequalities on graphs
- 3.7 Regions

4 Graphs and transformations

- 4.1 Cubic graphs
- 4.2 Reciprocal graphs
- 4.3 Points of intersection
- 4.4 Translating graphs
- 4.5 Stretching graphs
- 4.6 Transforming functions

5 Straight line graphs

- 5.1 $y = mx + c$
- 5.2 Equations of straight lines
- 5.3 Parallel and perpendicular lines
- 5.4 Length and area

6 Trigonometric ratios

- 6.1 The cosine rule
- 6.2 The sine rule
- 6.3 Areas of triangles
- 6.4 Solving triangle problems
- 6.5 Graphs of sine, cosine and tangent
- 6.6 Transforming trigonometric graphs

7 Radians

- 7.1 Radian measure
- 7.2 Arc length
- 7.3 Areas of sectors and segments

8 Differentiation

- 8.1 Gradients of curves
- 8.2 Finding the derivative
- 8.3 Differentiating x^n
- 8.4 Differentiating quadratics
- 8.5 Differentiating functions with two or more terms
- 8.6 Gradients, tangents and normals
- 8.7 Second order derivatives

9 Integration

9.1 Integrating x^n

9.2 Indefinite integrals

9.3 Finding functions

P2 — Pure Mathematics 2

RESEARCHED

1 Algebraic methods

- 1.1 Proof (deduction, exhaustion, counter-example)
- 1.2 Dividing polynomials
- 1.3 The factor theorem

2 Coordinate geometry in the (x, y) plane

- 2.1 Midpoints and perpendicular bisectors
- 2.2 Equation of a circle
- 2.3 Intersections of straight lines and circles
- 2.4 Tangent and chord properties
- 2.5 Circles and triangles

3 Exponentials and logarithms

- 3.1 Exponential functions
- 3.2 $y = e^x$
- 3.3 Exponential modelling
- 3.4 Logarithms
- 3.5 Laws of logarithms
- 3.6 Solving equations using logarithms
- 3.7 Working with natural logarithms
- 3.8 Logarithms and non-linear data

4 The binomial expansion

- 4.1 Pascal's triangle
- 4.2 Factorial notation
- 4.3 The binomial expansion
- 4.4 Solving binomial problems
- 4.5 Binomial estimation

5 Sequences and series

- 5.1 Arithmetic sequences
- 5.2 Arithmetic series
- 5.3 Geometric sequences
- 5.4 Geometric series
- 5.5 Sum to infinity
- 5.6 Sigma notation
- 5.7 Recurrence relations
- 5.8 Modelling with series

6 Trigonometric identities and equations

- 6.1 Angles in all four quadrants
- 6.2 Exact values of trigonometric ratios
- 6.3 Trigonometric identities
- 6.4 Simple trigonometric equations
- 6.5 Harder trigonometric equations
- 6.6 Equations and identities

7 Differentiation

- 7.1 Increasing and decreasing functions
- 7.2 Stationary points
- 7.3 Sketching gradient functions
- 7.4 Modelling with differentiation

8 Integration

- 8.1 Definite integrals
- 8.2 Areas under curves
- 8.3 Areas under the x-axis
- 8.4 Areas between curves and lines
- 8.5 The trapezium rule

P3 — Pure Mathematics 3

RESEARCHED

1 Algebraic methods

- 1.1 Simplifying algebraic fractions
- 1.2 Multiplying and dividing algebraic fractions
- 1.3 Adding and subtracting algebraic fractions
- 1.4 Dividing algebraic expressions

2 Functions and graphs

- 2.1 The modulus function
- 2.2 Functions and mappings
- 2.3 Composite functions
- 2.4 Inverse functions
- 2.5 $y = |f(x)|$ and $y = f(|x|)$
- 2.6 Combining transformations
- 2.7 Solving modulus problems

3 Trigonometric functions

- 3.1 Secant, cosecant and cotangent
- 3.2 Graphs of sec, cosec and cot
- 3.3 Using sec, cosec and cot
- 3.4 Trigonometric identities
- 3.5 Inverse trigonometric functions

4 Trigonometric addition formulae

- 4.1 Addition formulae
- 4.2 Double-angle formulae
- 4.3 Solving trigonometric equations
- 4.4 The form $R \cos(\theta \pm \alpha)$ / $R \sin(\theta \pm \alpha)$
- 4.5 Proving trigonometric identities

5 Exponentials and logarithms

- 5.1 e^x and $\ln x$
- 5.2 Derivatives of e^x and $\ln x$
- 5.3 Exponential modelling, growth and decay

6 Differentiation

- 6.1 The chain rule
- 6.2 The product rule
- 6.3 The quotient rule
- 6.4 Differentiating trigonometric functions
- 6.5 Implicit differentiation
- 6.6 Parametric differentiation
- 6.7 Rates of change

7 Integration

- 7.1 Standard integrals
- 7.2 Integration by substitution
- 7.3 Integration by parts
- 7.4 Integration using partial fractions
- 7.5 Areas and the trapezium rule

8 Numerical methods

- 8.1 Locating roots
- 8.2 Iteration
- 8.3 The Newton–Raphson method

P4 — Pure Mathematics 4

RESEARCHED

1 Proof

- 1.1 Proof by contradiction

2 Partial fractions

- 2.1 Partial fractions
- 2.2 Repeated factors
- 2.3 Improper fractions

3 Coordinate geometry in the (x, y) plane

- 3.1 Parametric equations
- 3.2 Using trigonometric identities
- 3.3 Curve sketching
- 3.4 Points of intersection

4 Binomial expansion

- 4.1 Expanding $(1 + x)^n$ for any rational n
- 4.2 Expanding $(a + bx)^n$
- 4.3 Using partial fractions

5 Differentiation

- 5.1 Parametric differentiation
- 5.2 Connected rates of change
- 5.3 Forming differential equations

6 Integration

- 6.1 Further integration techniques
- 6.2 Volumes of revolution
- 6.3 Solving differential equations (separating variables)

7 Vectors

- 7.1 3D coordinates
- 7.2 Vectors in three dimensions
- 7.3 The equation of a line in three dimensions
- 7.4 Points of intersection
- 7.5 The scalar product
- 7.6 Angles between lines

M1 — Mechanics 1

RESEARCHED

1 Mathematical models in mechanics

- 1.1 Constructing a model
- 1.2 Modelling assumptions
- 1.3 Quantities and units

2 Constant acceleration

- 2.1 Displacement–time graphs
- 2.2 Velocity–time graphs
- 2.3 Constant acceleration formulae (suvat)
- 2.4 Vertical motion under gravity

3 Vectors in mechanics

- 3.1 Working with vectors
- 3.2 Velocity as a vector
- 3.3 Solving problems with vectors
- 3.4 Forces given as vectors

4 Dynamics of a particle in a straight line

- 4.1 Force diagrams
- 4.2 Forces and acceleration (Newton's laws)
- 4.3 Motion in two dimensions
- 4.4 Connected particles
- 4.5 Pulleys

5 Forces and friction

- 5.1 Resolving forces
- 5.2 Inclined planes
- 5.3 Friction

6 Momentum and impulse

- 6.1 Momentum in one dimension
- 6.2 Impulse
- 6.3 Conservation of momentum

7 Statics of a particle

- 7.1 Static particles
- 7.2 Modelling with statics
- 7.3 Friction and static particles

8 Moments

- 8.1 Moments
- 8.2 Resultant moments
- 8.3 Equilibrium
- 8.4 Non-uniform rods

S1 — Statistics 1

RESEARCHED

1 Mathematical modelling

- 1.1 Mathematical models
- 1.2 Designing a model

2 Measures of location and spread

- 2.1 Measures of central tendency (mean, median, mode)
- 2.2 Quartiles and percentiles
- 2.3 Measures of spread
- 2.4 Variance and standard deviation
- 2.5 Coding

3 Representations of data

- 3.1 Histograms
- 3.2 Outliers
- 3.3 Box plots
- 3.4 Skewness
- 3.5 Comparing distributions

4 Probability

- 4.1 Understanding the vocabulary of probability
- 4.2 Venn diagrams
- 4.3 Mutually exclusive and independent events
- 4.4 Set notation
- 4.5 Conditional probability
- 4.6 Tree diagrams

5 Correlation and regression

- 5.1 Scatter diagrams and correlation
- 5.2 The product moment correlation coefficient
- 5.3 The least-squares regression line
- 5.4 Interpolation and extrapolation

6 Discrete random variables

- 6.1 Discrete random variables
- 6.2 The cumulative distribution function
- 6.3 Expected value $E(X)$
- 6.4 Variance $\text{Var}(X)$
- 6.5 Expected value and variance of a function of X
- 6.6 Solving problems involving random variables
- 6.7 The discrete uniform distribution

7 The Normal distribution

- 7.1 The normal distribution
- 7.2 Using tables to find probabilities
- 7.3 Finding the value of z given a probability
- 7.4 The standard normal distribution
- 7.5 Finding μ and σ

Part 2 — Further Mathematics

FP1 — Further Pure Mathematics 1

RESEARCHED

1 Complex numbers

- 1.1 Imaginary and complex numbers
- 1.2 Multiplying complex numbers
- 1.3 Complex conjugation
- 1.4 Argand diagrams
- 1.5 Modulus and argument
- 1.6 Modulus–argument form

2 Roots of quadratic equations

- 2.1 Roots of a quadratic equation
- 2.2 Solving cubic and quartic equations
- 2.3 Sums and products of roots

3 Numerical solutions of equations

- 3.1 Locating roots
- 3.2 Interval bisection
- 3.3 Linear interpolation
- 3.4 The Newton–Raphson method

4 Coordinate systems

- 4.1 Parametric equations
- 4.2 The parabola
- 4.3 The rectangular hyperbola
- 4.4 Tangents and normals
- 4.5 Loci

5 Matrices

- 5.1 Introduction to matrices
- 5.2 Matrix multiplication
- 5.3 Determinants
- 5.4 Inverting a 2×2 matrix

6 Transformations using matrices

- 6.1 Linear transformations in two dimensions
- 6.2 Reflections and rotations
- 6.3 Enlargements and stretches
- 6.4 Successive transformations

7 Series

- 7.1 Sums of natural numbers (Σr)
- 7.2 Sums of squares and cubes (Σr^2 , Σr^3)

8 Proof by induction

- 8.1 Proof by mathematical induction
- 8.2 Proving divisibility results
- 8.3 Recurrence relations
- 8.4 Matrix statements

FP2 — Further Pure Mathematics 2

RESEARCHED

1 Inequalities

- 1.1 Algebraic methods
- 1.2 Using graphs to solve inequalities
- 1.3 Modulus inequalities

2 Series

- 2.1 The method of differences

3 Complex numbers

- 3.1 Exponential form of a complex number
- 3.2 de Moivre's theorem
- 3.3 Trigonometric identities
- 3.4 Sums of series
- 3.5 nth roots of a complex number

4 Further Argand diagrams

- 4.1 Loci in the Argand diagram
- 4.2 Regions in the Argand diagram
- 4.3 Transformations of the complex plane

5 First order differential equations

- 5.1 Separating the variables
- 5.2 Exact equations
- 5.3 Integrating factors
- 5.4 Reducible first order equations

6 Second order differential equations

- 6.1 Homogeneous equations (auxiliary equation)
- 6.2 Non-homogeneous equations (particular integral)
- 6.3 Using boundary conditions
- 6.4 Reducible second order equations

7 Maclaurin and Taylor series

- 7.1 Maclaurin series
- 7.2 Taylor series
- 7.3 Series solutions of differential equations

8 Polar coordinates

- 8.1 Polar and Cartesian coordinates
- 8.2 Sketching polar curves
- 8.3 Area enclosed by a polar curve
- 8.4 Tangents to polar curves

FP3 — Further Pure Mathematics 3

RESEARCHED

1 Hyperbolic functions

- 1.1 Introduction to hyperbolic functions
- 1.2 Inverse hyperbolic functions
- 1.3 Identities and equations

2 Further coordinate systems

- 2.1 Cartesian and parametric equations of ellipses and hyperbolas
- 2.2 Eccentricity, foci and directrices
- 2.3 Tangents and normals
- 2.4 Loci

3 Differentiation

- 3.1 Differentiating hyperbolic functions
- 3.2 Differentiating inverse trigonometric and hyperbolic functions

4 Integration

- 4.1 Integration of hyperbolic functions
- 4.2 Inverse trig and hyperbolic forms
- 4.3 Integration using hyperbolic and trigonometric substitution
- 4.4 Reduction formulae
- 4.5 Arc length and area of a surface of revolution

5 Vectors

- 5.1 The vector (cross) product
- 5.2 The scalar triple product
- 5.3 Straight lines
- 5.4 Planes
- 5.5 Distances and angles

6 Further matrix algebra

- 6.1 Determinants of 3×3 matrices
- 6.2 The inverse of a 3×3 matrix
- 6.3 Solving systems of linear equations
- 6.4 Eigenvalues and eigenvectors
- 6.5 Diagonalising a matrix

D1 — Decision Mathematics 1

RESEARCHED

1 Algorithms

- 1.1 Using and understanding algorithms
- 1.2 Flow charts
- 1.3 Bubble sort
- 1.4 Quick sort
- 1.5 Bin-packing algorithms
- 1.6 Order of an algorithm

2 Graphs and networks

- 2.1 Modelling with graphs
- 2.2 Graph theory terminology
- 2.3 Special types of graph
- 2.4 Representing graphs and networks using matrices

3 Algorithms on graphs

- 3.1 Kruskal's algorithm
- 3.2 Prim's algorithm
- 3.3 Prim's algorithm from a matrix
- 3.4 Dijkstra's algorithm (shortest path)

4 Route inspection

- 4.1 The route inspection (Chinese postman) problem

5 The travelling salesman problem

- 5.1 The classical and practical problems
- 5.2 Using a minimum spanning tree to find bounds
- 5.3 The nearest neighbour algorithm

6 Critical path analysis

- 6.1 Precedence tables
- 6.2 Activity networks and dummy activities
- 6.3 Early and late event times
- 6.4 Critical activities and float
- 6.5 Gantt (cascade) charts and scheduling

7 Linear programming

- 7.1 Formulating linear programming problems
- 7.2 Graphical methods
- 7.3 The objective line (ruler) method
- 7.4 The vertex method
- 7.5 Integer solutions

M2 — Mechanics 2

RESEARCHED

1 Kinematics

- 1.1 Motion with variable acceleration (calculus)
- 1.2 Motion in two dimensions with vectors
- 1.3 Projectiles
- 1.4 The projectile formulae

2 Centres of mass

- 2.1 Centre of mass of a system of particles
- 2.2 Centre of mass of a uniform lamina
- 2.3 Standard uniform laminae and frameworks
- 2.4 Laminae in equilibrium (suspended)

3 Work, energy and power

- 3.1 Work done by a force
- 3.2 Kinetic and potential energy
- 3.3 Conservation of mechanical energy and the work–energy principle
- 3.4 Power

4 Impulses and collisions

- 4.1 Momentum as a vector
- 4.2 Direct impact and Newton's law of restitution
- 4.3 Direct collisions with a smooth plane
- 4.4 Loss of kinetic energy
- 4.5 Successive direct impacts

5 Statics of rigid bodies

- 5.1 Static rigid bodies
- 5.2 Moments and equilibrium
- 5.3 Rigid bodies with friction

M2 chapter split (work/energy/power and collisions) can run across an extra chapter in some editions — worth a quick check against the contents page.

S2 — Statistics 2

RESEARCHED

1 The binomial distribution

- 1.1 The binomial distribution
- 1.2 Using binomial tables
- 1.3 Mean and variance

2 The Poisson distribution

- 2.1 The Poisson distribution
- 2.2 Using Poisson tables
- 2.3 Mean and variance

3 Approximations

- 3.1 The Poisson approximation to the binomial
- 3.2 The normal approximation to the binomial
- 3.3 The normal approximation to the Poisson

4 Continuous random variables

- 4.1 The probability density function (p.d.f.)
- 4.2 The cumulative distribution function (c.d.f.)
- 4.3 Mean and variance
- 4.4 Mode, median and quartiles

5 The continuous uniform distribution

- 5.1 The continuous uniform (rectangular) distribution
- 5.2 Modelling with the continuous uniform distribution

6 Sampling and sampling distributions

- 6.1 Populations and samples
- 6.2 Sampling methods
- 6.3 Sampling distributions

7 Hypothesis testing

- 7.1 Hypotheses and test statistics
- 7.2 Tests for a binomial proportion
- 7.3 Tests for a Poisson mean
- 7.4 Critical regions and significance levels

Unit-to-qualification map

Qualification	Pure units	Applied units
A Level Mathematics	P1, P2, P3, P4	M1, S1
A Level Further Mathematics	FP1, FP2, FP3	D1, M2, S2

Edexcel (Pearson) International Advanced Level — Mathematics & Further Mathematics (2018 spec). Chapter structures verified against the Pearson IAL student books; P1 sub-sections verified against the contents page; other sub-sections are best-effort within the verified chapters. Prepared for Cranleigh Abu Dhabi Y12/Y13.